Université Hassan 1er
Faculté des Sciences et Techniques Settat
Génie Electrique & Génie Mécanique

SYSTÈME D'INFORMATION ET TRANSFORMATION DIGITALE

SMART PARKING PROJECT REPORT



2023-2024

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Words cannot fully express our profound gratitude to Professor Karim Tahiry for his exceptional dedication to teaching us about IoT and for his invaluable leadership in guiding our IoT smart parking project. Throughout this journey, Professor Tahiry's unwavering support, expertise, and commitment have shaped our understanding and accomplishments in this complex and dynamic field.

Professor Tahiry's passion for IoT and his tireless efforts to ensure our learning experience was both enriching and challenging have left an indelible mark on our academic and professional growth. His insightful guidance, patient mentorship, and willingness to go above and beyond to assist us have truly made a difference.

Thank you, Professor Tahiry, for your outstanding contributions to our education and for inspiring us to reach new heights in the world of IoT. Your passion, expertise, and unwavering support have been the cornerstone of our success, and we are truly honored to have been under your mentorship.

GENERAL INTRODUCTION :

In anticipation of hosting major global sporting events like the Africa Cup of Nations and the World Cup, Morocco stands on the precipice of a transformative opportunity to showcase its prowess on the international stage. As the country gears up to welcome a significant influx of visitors, particularly in urban hubs like Casablanca, the need to effectively manage transportation and alleviate traffic congestion becomes increasingly pressing.

The sprawling metropolis of Casablanca, already grappling with urban mobility challenges, faces the daunting task of accommodating the surge in vehicular and pedestrian traffic expected during these high-profile events. The potential for gridlock and logistical bottlenecks looms large, threatening to mar the visitor experience and undermine Morocco's image as a capable host.

To preemptively address these challenges and ensure the seamless flow of people and vehicles, innovative solutions rooted in cutting-edge technology are being explored. Among these solutions, IoT-based parking systems emerge as a beacon of promise, offering the potential to revolutionize parking management and streamline traffic flow.

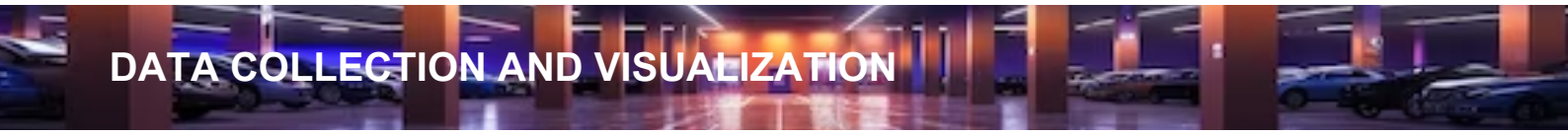
By harnessing IoT technology, these smart parking systems enable real-time monitoring of parking availability, guiding motorists to vacant spaces with pinpoint accuracy. Moreover, integrated management platforms provide centralized oversight, allowing for agile response to changing traffic conditions and dynamic event schedules.

WHY SMART PARKING SOLUTION?

Implementing such solutions optimizes parking facilities, minimizes congestion, and enhances the overall visitor experience. With reduced waiting times and improved accessibility, attendees can navigate the urban landscape with greater ease, fostering a sense of comfort and convenience that reflects positively on Morocco's hosting capabilities.

Beyond the immediate benefits to event attendees, the successful deployment of IoT-based parking systems serves as a testament to Morocco's commitment to innovation and sustainability. By embracing technology-driven solutions, the country demonstrates its proactive stance in addressing urban mobility challenges while advancing its reputation as a forward-thinking and progressive nation.

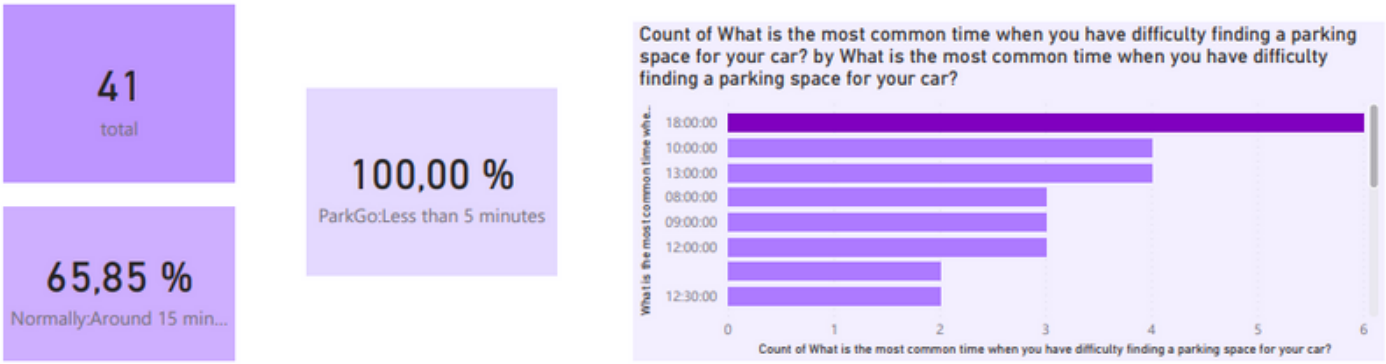
In this context, the forthcoming sporting events represent not only a showcase of athletic prowess but also an opportunity for Morocco to demonstrate its resilience, adaptability, and capacity for transformation on the global stage. As the world's gaze turns towards Casablanca and beyond, the implementation of IoT-based parking systems stands as a testament to Morocco's readiness to rise to the occasion and leave a lasting impression as a host of distinction.

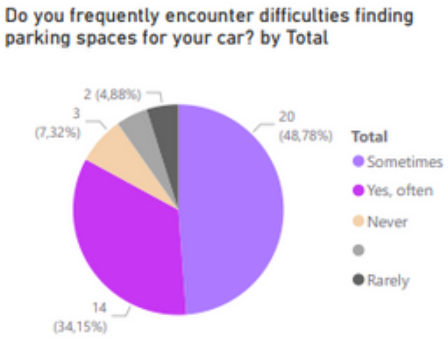


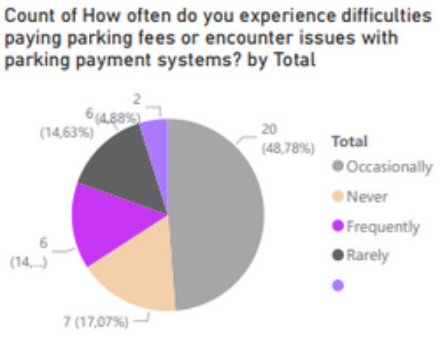
DATA COLLECTION AND VISUALIZATION

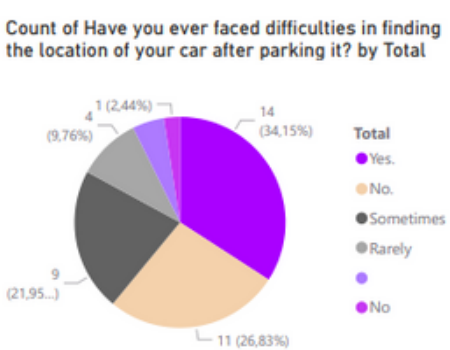
For the data collection phase of our project, we implemented a Google Form to gather valuable insights from drivers spanning across Morocco. This method allowed us to directly engage with our target audience and gather diverse perspectives on parking issues and preferences.

Subsequently, we utilized Power BI, a powerful business analytics tool, to visualize the data gathered from the Google Form. Through interactive dashboards and visualizations, we were able to delve deeper into the utility and feasibility of the Smart Parking project.







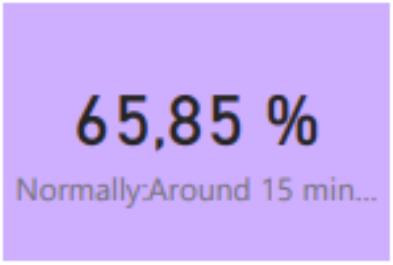




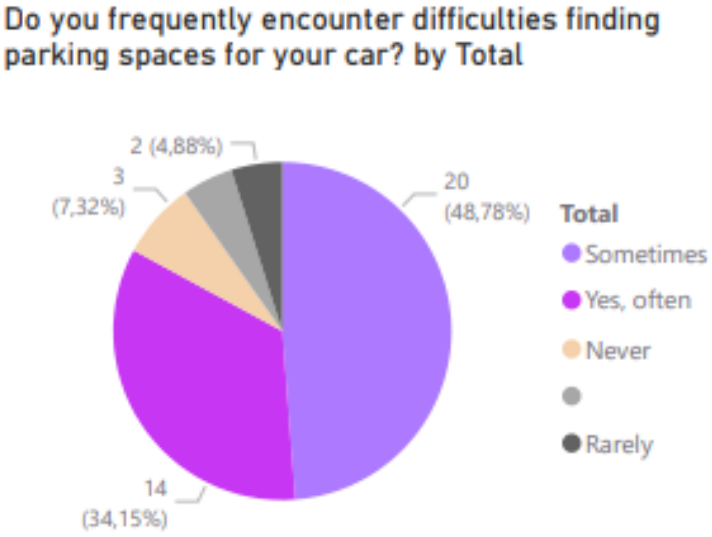
UNDERSTANDING PARKGO'S IMPACT

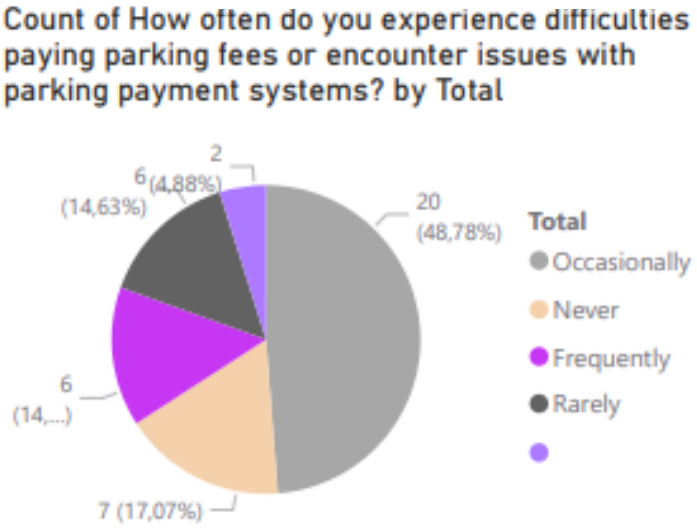
The analysis provided by Power BI shed light on the reasons why ParkGo holds significant value for drivers. It became evident that ParkGo has the potential to address a myriad of parking challenges faced by individuals across various regions of Morocco. By visualizing the data, we gained a comprehensive understanding of the project's potential impact.







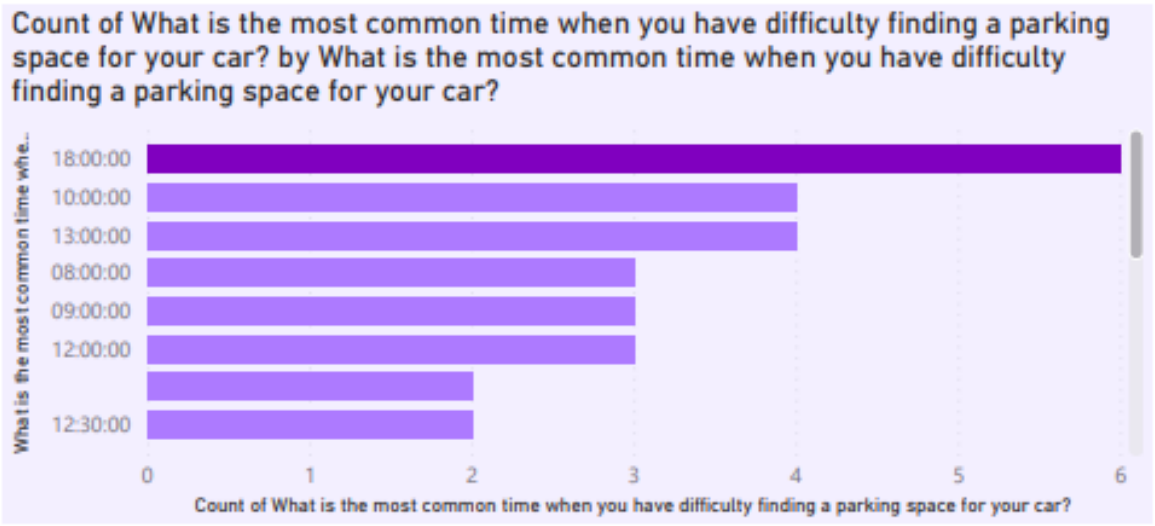


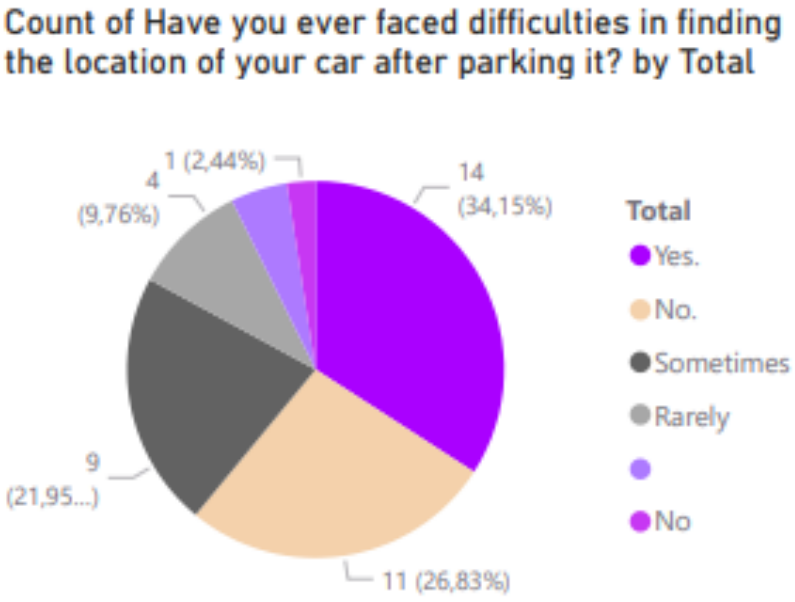




DATA-DRIVEN DECISION-MAKING

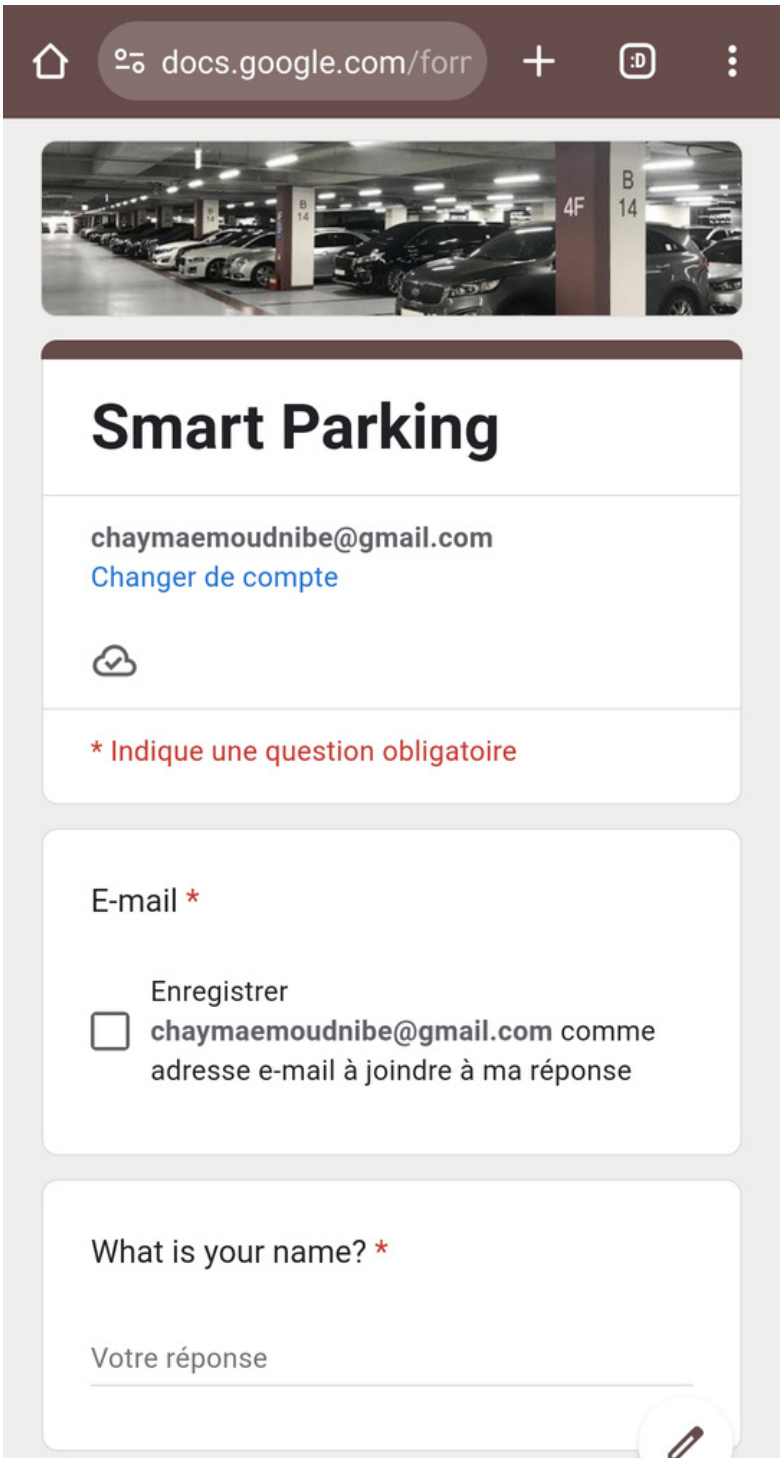
Moreover, this data-driven approach empowered us to make informed decisions throughout the project lifecycle. The insights derived from the data not only guided our strategy but also served as a foundation for making recommendations. By leveraging the analytics provided by Power BI, we were able to identify key trends and patterns, facilitating proactive measures to tackle parking problems efficiently.





OUR SURVEY TO COLLECT DATA:

https://docs.google.com/forms/d/1P6xefZQoR43RaLjkpe4VeR0ZXAw3qXVsBpRYoU81K2E/viewform?edit_requested=true#settings





OUTILS:

Power BI is a powerful business analytics tool developed by Microsoft. It enables users to visualize and analyze data through interactive reports and dashboards. With Power BI, users can connect to various data sources, transform raw data into meaningful insights, and make data-driven decisions.

Google Form is a free tool provided by Google for creating online surveys and forms. It allows users to easily gather information and feedback from respondents. With Google Form, users can create customized forms with various question types, share them with participants, and collect and analyze responses in a structured format.

CHAPTER 1: OVERVIEW OF SMART PARKING

1. INTRODUCTION

Car parking is a topic that has always been prevalent because it affects the mobility of most individuals. Drivers always need to park to reach their destinations, and a solution of an automated car parking system is proposed to solve or reduce the problems encountered in conventional parking lots.

2. DEFINITIONS

2.1 Parking

Car parking refers to a collection of registered vehicles parked for a specified period within an enclosed area. Parking lots are found in various locations (airports, hospitals, major markets, etc.) and come in different types (outdoor, multi-level, and automated parking: in a tower and doubling the space...).

2.2 Smart Parking

An ideal solution for today's parking lots and traffic problems in cities. An automated parking system offers maximum efficiency, convenience, safety, and reliability. Powered by electricity, automated parking systems automatically park and retrieve cars, using a robotic system for vehicle storage, generating less CO2 than conventional multi-level parking lots.

3. Parking Problems

Cities face a major parking problem, with the most popular areas experiencing ever-increasing demand for parking. Parking is not only for local residents but for everyone, and the duration of parking is not specific, which causes problems. Despite the presence of parking lots, many accidents and issues occur among drivers. To reduce these problems, one solution is smart parking. It reduces issues by helping tourists, employees, or anyone unfamiliar with the city to easily and reliably reserve a parking space for their vehicle.

4. Smart Parking Vs Traditional Parking



5. Types of smart parking systems

There are several types of solutions in this category, based on the devices used.

- **Smart parking solution powered by ground sensors**. Ground sensors are not the latest innovations. In fact, they've been in use for several decades with little change. They use radar to detect when a car occupies a parking spot.
- **Counter system**. Such solutions usually rely on raisable gates. They count the number of cars that have entered and exited. A system of this type is a popular choice for an enclosed parking garage. These systems have been widely adopted but have disadvantages if they're not accompanied by additional sensors. They can't demonstrate smart parking maps with available places.
- **Camera-based systems or systems enriched with overhead parking sensors**. Smart detectors of this kind can view a parking area from above, each covering more than one parking space at the same time.



6. Benefits of a smart parking system

When considering the feasibility of implementing a smart parking system, it's worth learning about the core advantages.

1. **Less stress for drivers.** Trying to find an available parking space can become an exercise in frustration. But smart detectors and cameras installed at a parking lot and connected to a cloud server can solve the problem. If you can readily see that all spaces are occupied, you can just look for another parking area.
2. **Fuel and time savings.** Apps give car owners a clear understanding of where they can find a parking space and the chance to book it in advance. This means less wasted time and fuel spent driving around looking for a parking spot.
3. **Informed parking management decisions.** The possibility of monitoring the situation at a parking lot remotely is also beneficial for operators, who can build more efficient business strategies and make better decisions about whether to close a parking garage or open a new one in a particular zone. This means city authorities and businesses can make smarter investments in organizing parking areas and optimize their budget allocations.



4. **Higher revenues for parking operators.** Real-time data analytics can be used not only for better planning and service enhancement but also for introducing dynamic pricing. In this case, prices can be changed depending on ongoing demand, which will help maximize profits.
5. **Lower carbon footprint.** Providing drivers with relevant information on parking availability will minimize the number of cars circling around a garage, thus reducing pointless carbon emissions.
6. **Reduced traffic jams.** This benefit is closely related to the previous one. When drivers clearly know where they can leave their vehicles, they don't need to wait in lines to get out of a garage after an unsuccessful attempt at finding a parking space.
7. **Easy payments.** A smart parking system can also accept payments via a website. The transaction is simple and quick, enhancing the experience for drivers.
8. **Higher safety and security**. A smart parking system enriched with cameras helps operators monitor what is happening in the parking lot. It is possible to know who is arriving, by identifying the license plate, and when they arrive and depart.

Conclusion

Smart parking stands as a pivotal solution in modern urban planning, offering a multifaceted approach to alleviate congestion and enhance urban mobility. By harnessing advanced technologies and innovative systems, smart parking solutions address the pressing challenges faced by cities worldwide.
In this chapter, we provided an overview of smart parking, its various types, definition, and its advantages.

CHAPTER 2: ELECTRONIC COMPONENT

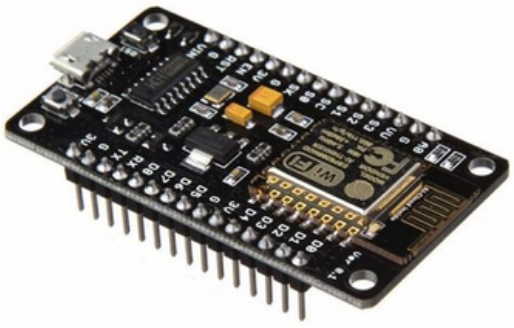
1. Introduction:

The choice of materials is a fundamental and highly complex task, and to successfully address it, we established a set of criteria that needed to be satisfied for our selection to be successful. The imposed criteria are as follows:

- Material level: mechanical, aesthetic, thermal, electrical, economic, environmental, and physical characteristics.
- Process level: volume, mass, geometry, series size, economic, and environmental characteristics.

A) NODEMCU(ESP8266)

The NodeMCU is an open-source firmware and development kit that helps you to Prototype your product within a few Lua script lines. ESP8266 has powerful on-board processing and storage capabilities that allow it to be integrated with the sensors specific devices through its GPIOs with minimal development up-front and minimal loading during runtime.



ESP8266 Specifications
This is based on the ESP-12 module, which we discuss below

Search this file…

	ESP8266	DESCRIPTION
1	Core	1
2	Arquitecture	32 bits
3	Clock	Xtensa LX106 80-160MHz
4	WiFi	IEEE802.11 b/g/n support for WPA and WPA2
5	Bluetooth	No
6	RAM	160KB - 64KB Instruction - 96KB Data
7	Flash	Extern QSPI - 512KB A 4MB
8	GPIO	16
9	DAC	0
10	ADC	1
11	Interfaces	SPI-I2C-UART-I2S
12		

ESP8266.csv hosted with ♥ by GitHub view raw

ESP8266 Functions
ESP8266 has many applications when it comes to the IoT. Here are just some of the functions the chip is used for:

- Networking: The module's Wi-Fi antenna enables embedded devices to connect to routers and transmit data
- Data Processing: Includes processing basic inputs from analog and digital sensors for far more complex calculations with an RTOS or Non-OS SDK
- P2P Connectivity: Create direct communication between ESPs and other devices using IoT P2P connectivity
- Web Server: Access pages written in HTML or development languages.

ESP8266 Applications
The ESP8266 modules are commonly found in the following IoT devices:

- Smart security devices, including surveillance cameras and smart locks
- Smart energy devices, including HVACs and thermostats
- Smart industrial devices, including Programmable Logic Controllers (PLCs)
- Smart medical devices, including wearable health monitor.

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Chip versus Modules versus Development Boards

As discussed above, the ESP8266 is just the name of the chip. There are essentially three formats you can buy this in:

- ESP8266 Chip: This is the basic chip manufactured by Espressif, which comes unshielded and needs to be soldered onto a module. This is unsuitable for most users, apart from perhaps volume device manufacturers that can factor this into the production process under the unit cost of a module.

- ESP8266 Modules: These are the surface-mountable modules that contain the chip, which are ready to be mounted onto an MCU, produced by Espressif, Ai-Thinker and certain other manufacturers. They are usually shielded and pre-approved by the FCC for use. This means they’re a good option for device manufacturers looking to scale production.
- ESP8266 Development Boards: These are the complete IoT MCU development boards that have the modules preinstalled. They’re used for developers and manufacturers to create prototypes during the design stage, before they start production. Development boards are produced by several different manufacturers and the specifications differ between models. Some core specifications to be aware of when assessing ESP8266 IoT development board options include:
  - GPIO pins
  - ADC pins
  - Wi-Fi antennas
  - LEDs
  - Shielding*
  - Flash Memory

*Many international markets require shielded Wi-Fi devices, as Wi-Fi produces considerable Radio Frequency Interference (RFI), and shielding minimizes this interference. This should, therefore, be a key consideration for all developers and embedded-device manufacturers.

C) SERVO MOTOR

Servo Motor is a rotator device that allows the control of linear as well as angular motion. A Servo Motor is used for the opening and closing of the gate. Servo drive sends electrical signals to the Servo Motor for producing motion.

Types of Servo Motors

There are various types of servo motors, differing in size, weight, and torque (force). The photograph below shows a very typical servo motor in modeling: the Futaba S3003. A little further down the article, we will use another servo motor commonly referred to as a "9-gram servo motor" due to its electrical consumption.



a. 9-gram servo motor b.Servomotor Futuba S3003

Figure : Types of servo motors

Operating Principle of a Servo Motor

Most servo motors are controlled through a three-wire electrical cable that powers the motor and transmits position commands in the form of a pulse width modulation (PWM) signal. This means that the duration of the pulses determines the absolute angle of the output shaft and thus the position of the servo motor's control arm. The signal is repeated periodically, typically every 20 milliseconds, allowing the electronics to continuously control and correct the angular position of the output shaft, which is measured by the potentiometer.

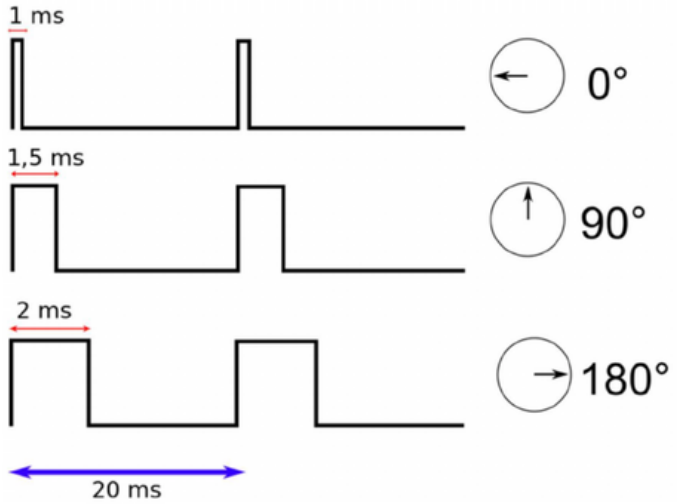
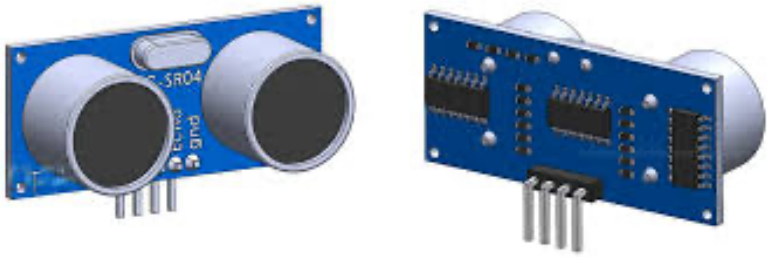


Figure : Types of servo motors

B) ULTRASONIC SENSOR

An ultrasonic sensor is an electronic device that is use to measure the distance of an object by emitting ultrasonic sound waves and converts the reflected sound waves into an electrical signal. The speed of Ultrasonic waves are faster than the speed of an audible sound.



How Does an Ultrasonic Sensor Work?

Moving from theory to reality, an ultrasonic sensor requires two parts, both a transmitter and a receiver. In the most standard configuration, these are placed side-by-side as close together as reasonably possible. With the receiver close to the transmitter, sound travels in a straighter line from the transmitter to the detected object and back to the receiver, yielding smaller errors in the measurements. There are also ultrasonic transceivers where the transmitter and receiver functions are integrated into a single unit, minimizing error as much as physically possible while also significantly reducing the PCB footprint.

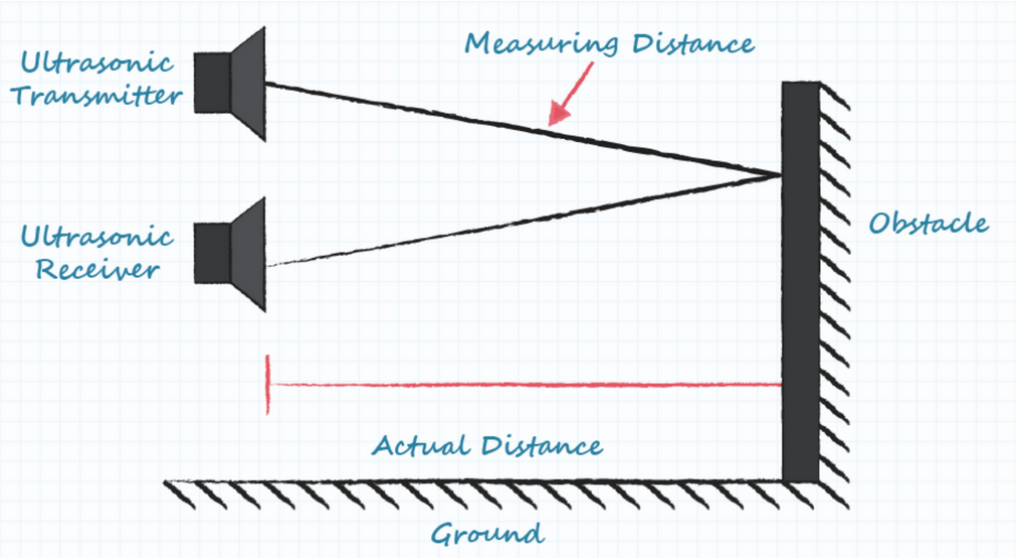


Figure : Basic operation of an ultrasonic transmitter and receiver pair

Strengths of Ultrasonic Sensors:

1. Non-contact Measurement: Ultrasonic sensors measure distance without physical contact, making them suitable for various applications where contact sensors are not feasible.
2. High Accuracy: They offer high accuracy in distance measurement, making them suitable for precision applications.
3. Wide Range: Ultrasonic sensors can detect objects over a wide range, from a few centimeters to several meters, depending on the sensor's specifications.
4. Versatility: They can be used in diverse environments and conditions, including indoor and outdoor settings.
5. Immunity to Color and Transparency: Ultrasonic sensors can detect objects regardless of their color or transparency, unlike optical sensors.
6. Reliability: They are generally robust and reliable, with minimal susceptibility to environmental factors such as dust, light, or humidity.

Weaknesses of Ultrasonic Sensors:

1. Limited Resolution: Ultrasonic sensors may have limited resolution compared to some other distance measurement technologies.
2. Sensitivity to Temperature Changes: Extreme temperature variations can affect the speed of sound in air, potentially impacting the accuracy of ultrasonic measurements.

3. Beam Spread: Ultrasonic waves spread out as they travel, leading to a wider detection area and potential interference from nearby objects.
4. Reflective Surfaces: Highly reflective surfaces can cause echoes, leading to inaccuracies or false readings in distance measurement.
5. Limited Detection Range in Certain Conditions: Ultrasonic sensors may struggle to detect objects in conditions with heavy fog, rain, or dust, which can absorb or scatter ultrasonic waves.
6. Interference from Acoustic Noise: Background noise, such as loud machinery or other ultrasonic sensors nearby, can interfere with the sensor's operation, affecting accuracy.

Conclusion

The selection of materials was the most delicate phase in our project. It not only involved purely technical aspects meeting functional requirements but also addressed user preferences within a specific market context. Overall, the choice of materials was analyzed from the perspective of materials engineering, response time, not to mention the aesthetic aspect (industrial design).

CHAPTER3: COMPUTER TOOLS

1. Introduction:

The Arduino board with the ultrasonic sensor, the Servo Motor, and then the LCD display are fully programmed by the computer using software. For our project, we will use the Arduino 1.8.13 programming software.

2. Arduino IDE

a. Overview:

- The Arduino IDE is a software development environment specifically designed for programming the Arduino board, offering a wide range of features aimed at simplifying the development process of embedded projects.
- The Arduino language draws inspiration from several programming languages, giving it familiarity to developers from various backgrounds. This versatility makes learning and adapting to the Arduino language easier for users.
- The Arduino language imposes a particular structure that reflects the requirements of embedded computing, ensuring efficient use of the limited resources available on the Arduino board.
- The program written in the Arduino environment is interpreted by the microcontroller sequentially, from top to bottom, ensuring consistent execution of instructions in the order defined by the developer.

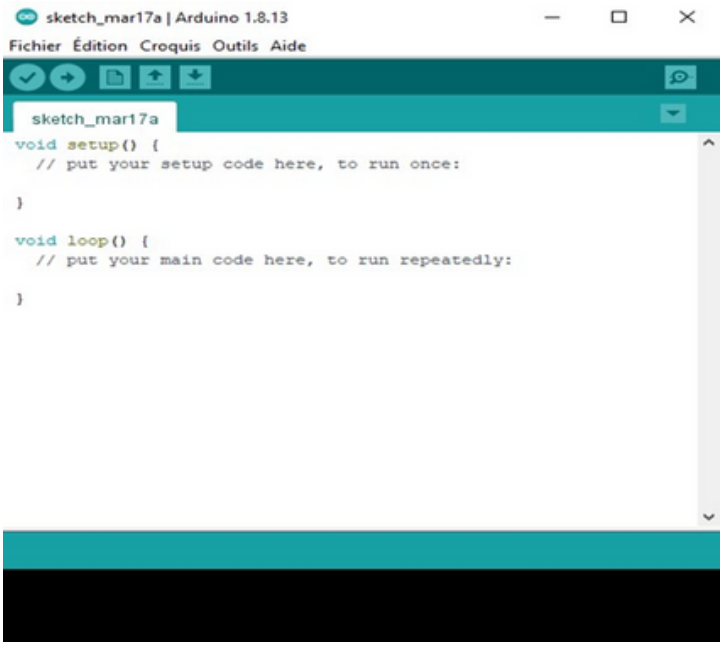
- The minimal structure of an Arduino program consists of three main parts:

- Header: This section includes variable declarations, constants, and indication of the use of necessary libraries for the project.
- Setup: This part of the code is executed only once at the startup of the Arduino microcontroller. It is used to perform tasks such as initializing pins as input or output, activating communication ports (such as MIDI, serial port, or I2C), etc.
- Loop: The Loop section is executed repeatedly after initialization. This is where the main functions of the program are performed repetitively.

- In addition to this minimal structure, additional elements can be added:

- Subroutines or routines: These parts of the code can be called at any time within the main loop. They are useful for executing repetitive pieces of code or for organizing the program in a modular way.

- Callbacks: These are special functions automatically called by certain libraries or events. Callbacks are used to handle interrupts, asynchronous events, or interactions with external devices.



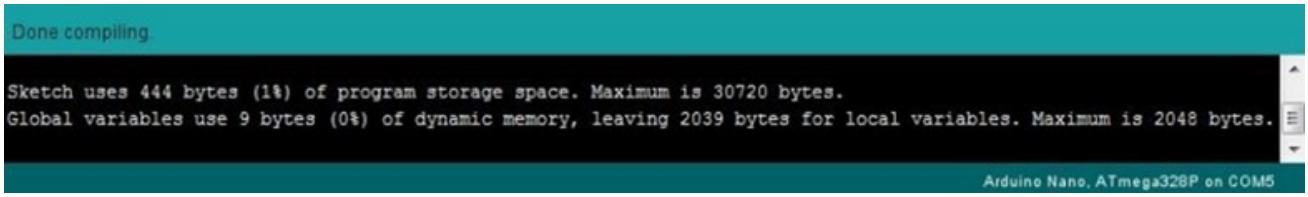
b. Programming Language:

The programming language used in the Arduino IDE development environment is C/C++. Arduino programs typically consist of the following sections:

- #include: Declaration of libraries to include.
- const ...: Declaration of constants.
- int ...: Declaration of global variables.
- void setup(): Initialization function executed after a restart. It is used to initialize elements that will not change during the program's execution.
- void loop(): Main function that runs in a loop. Your program executes here. Like in classic C, you can declare your own functions after this one.

c. Information Zone:

This black block contains compilation details and information, notably including a systematic summary of memory resource consumption.



d.Processing our Arduino Program:

i. Used Libraries:

- ESP8266WiFi.h: This library provides support for the ESP8266 WiFi module, enabling it to connect to a Wi-Fi network and communicate over the internet. It includes functions for configuring and managing Wi-Fi connections, as well as sending and receiving data over Wi-Fi. This library is commonly utilized in IoT (Internet of Things) projects involving the ESP8266 module. In scenarios like the one presented here, where the Arduino board is equipped with an ESP8266 chip for wireless communication capabilities, ESP8266WiFi.h facilitates seamless integration with Wi-Fi networks, enabling remote communication and control.

- **Servo.h:** Servo.h is a library that facilitates the control of servo motors. Servo motors are frequently employed in projects requiring precise control of angular position, such as robotic arms, motorized doors, or in the case presented here, a parking barrier. This library simplifies servo motor control by offering functions to attach a servo motor to a specific pin, set its position, and detach it when not in use. In this code, Servo.h is employed to regulate the position of the parking barrier by adjusting the angle of the servo motor.

- **LiquidCrystal.h:** LiquidCrystal.h is a library used for controlling LCD (Liquid Crystal Display) modules, particularly those based on the Hitachi HD44780 controller or compatible controllers. This library simplifies the process of interfacing with and controlling LCD displays in Arduino projects. It provides functions for initializing the LCD, displaying text, moving the cursor, and other display-related operations. By including LiquidCrystal.h in your Arduino code, you can easily incorporate LCD displays into your projects and enhance user interfaces with visual feedback.

```
#include <ESP8266WiFi.h>
#include <Servo.h>
#include<LiquidCrystal.h>
```

3.The developed program:

a.functionalities:

Pin Definitions:

- triggerPin, echoPin, RledPin, GledPin, and YledPin are defined to represent the pins connected to the HC-SR04 ultrasonic sensor and LEDs respectively. These pins are configured to control the sensor and the LEDs.
- sensorPin is defined as an analog input pin for potential future use.

Variable Declarations:

- duration and distance are declared to store the duration of the echo signal and the calculated distance measured by the ultrasonic sensor.

Servo Initialization:

- An instance of the Servo class named barrierServo is created to control the servo motor responsible for the parking barrier.

Setup Function:

- Serial communication is initialized at a baud rate of 9600.
- The yellow LED pin is configured as an output and turned on.
- Pins are configured as either inputs or outputs depending on their usage.
- The servo motor is attached to its corresponding pin.

Loop Function:

- A 10 µs pulse is sent to the trigger pin of the ultrasonic sensor to initiate distance measurement.
- The duration of the echo signal is read using the pulseIn function.
- The distance is calculated based on the duration of the echo signal.
- The measured distance is printed to the serial monitor.
- LEDs are controlled based on the measured distance:
- If the distance is less than 8 cm, the green LED is turned on and the red LED is turned off.Otherwise, the red LED is turned on and the green LED is turned off.
- If the distance is less than 10 cm, the barrier is opened for 5 seconds using the openBarrier function, followed by closing it using the closeBarrier function.

Barrier Control Functions:

- openBarrier function opens the barrier by setting the servo to a specific angle.
- closeBarrier function closes the barrier by setting the servo to another angle.

Delay:

- A delay of 500 milliseconds is added between each measurement to prevent rapid iteration.

b.The program:

```
#include <ESP8266WiFi.h>
#include <Servo.h>
#include<LiquidCrystal.h>

// Pins for HC-SR04 ultrasonic sensor and LEDs
const int triggerPin = D1;  // Trigger pin (TRIGGER)
const int echoPin = D2;     // Echo pin (ECHO)
const int RledPin = D3;     // Red LED pin
const int GledPin = D4;     // Green LED pin
const int YledPin = D6;     // Yellow LED pin

// Define sensor pin for analog input
#define sensorPin A0

// Variables declaration
long duration;
int distance;
Servo barrierServo;

void setup() {
 Serial.begin(9600); // Initialize serial communication
 // Configure the yellow LED pin as output
 pinMode(YledPin, OUTPUT);

 // Turn on the yellow LED
 digitalWrite(YledPin, HIGH);

 // Configure pins
 pinMode(triggerPin, OUTPUT);
 pinMode(echoPin, INPUT);
 pinMode(RledPin, OUTPUT); // Declare LED pins as output
 pinMode(GledPin, OUTPUT);
 pinMode(YledPin, OUTPUT);

 barrierServo.attach(D3);
}
```

```
void loop() {
  // Send a 10 µs pulse on the trigger pin
  digitalWrite(triggerPin, LOW);
  delayMicroseconds(2);
  digitalWrite(triggerPin, HIGH);
  delayMicroseconds(10);
  digitalWrite(triggerPin, LOW);

  // Read the duration of the echo signal
  duration = pulseIn(echoPin, HIGH);

  // Calculate distance in centimeters
  distance = duration * 0.034 / 2;

  // Print the measured distance
  Serial.print("Distance: ");
  Serial.print(distance);
  Serial.println(" cm");
// Control LEDs based on measured distance
 if (distance < 8 ) {
 digitalWrite(GledPin, HIGH); // Turn on green LED if distance is less than 8 cm
 digitalWrite(RledPin, LOW); // Turn off red LED
 digitalWrite(YledPin, HIGH); // Turn off yellow LED
 Serial.println("Green LED ON");
 } else {
 digitalWrite(RledPin, HIGH); // Turn on red LED if distance is 15 cm or more
 digitalWrite(GledPin, LOW); // Turn off green LED
 digitalWrite(YledPin, HIGH); // Turn off yellow LED
 Serial.println("Red LED ON");
 }
```

```
 // Control barrier based on measured distance
 if (distance < 10) {
 openBarrier();
 delay(5000);
 closeBarrier();
 }

 delay(500); // Delay between each measurement
}

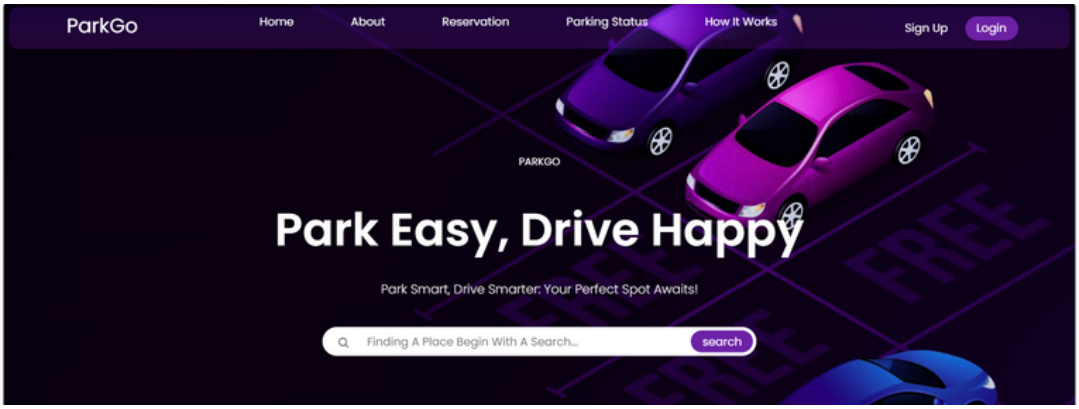
void openBarrier() {
 barrierServo.write(90); // Open position
 Serial.println("Barrier Opened");
 delay(1000);
}

void closeBarrier() {
 barrierServo.write(0); // Closed position
 Serial.println("Barrier Closed");
 delay(1000);
```

3.OVERVIEW OF THE WEBSITE:

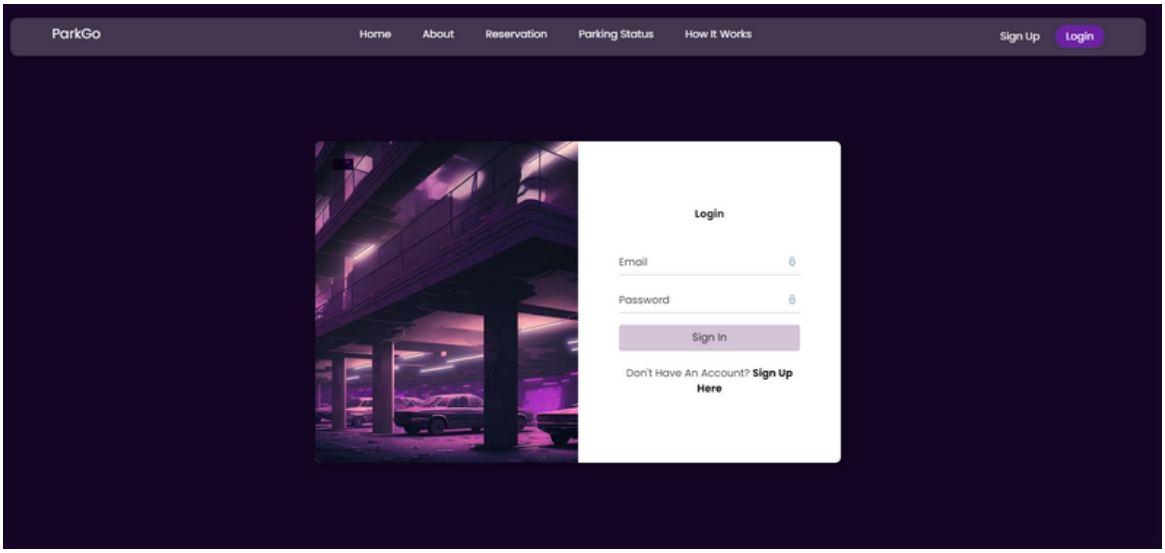
1. Interface Design:

The interface design of our smart parking website serves as the first point of contact for users, setting the tone for their interaction with the platform. Our interface features a modern and intuitive design aesthetic, characterized by clean layouts, vibrant colors, and clear typography. The interface is visually appealing and user-friendly, creating a welcoming environment for users to explore and interact with our smart parking system.



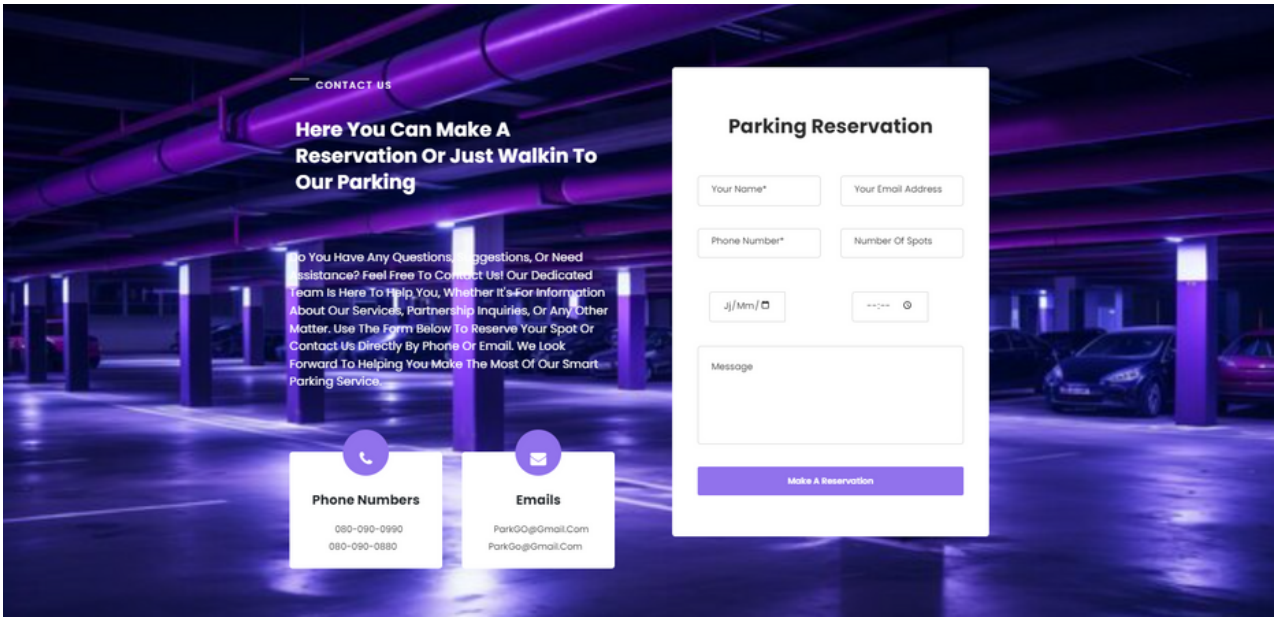
2. Sign-Up Feature:

The sign-up feature of our smart parking website plays a crucial role in providing personalized services to users and facilitating access to the reservation system. Users are required to register on the platform to access advanced features, including booking parking spots in advance. The registration process is streamlined and user-friendly, requiring users to provide essential information such as name, email address, and password. To ensure the security of user data, robust security measures such as encryption and secure authentication protocols have been implemented.



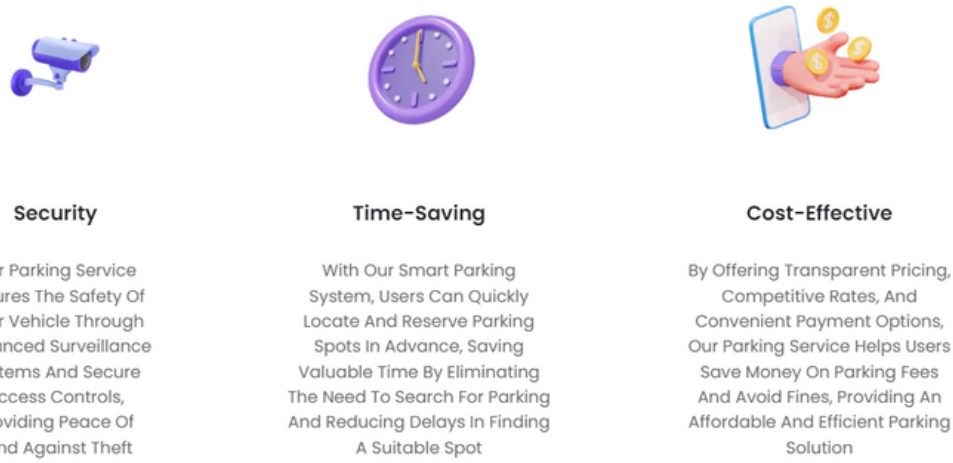
3. Reservation System:

Our reservation system enables users to conveniently book parking spots in advance, enhancing the efficiency of parking space management. Through the reservation system, users can select their preferred parking spot, specify the date and time of arrival, and make secure online payments. This functionality streamlines the parking process, reducing congestion and minimizing the time spent searching for available parking spaces.



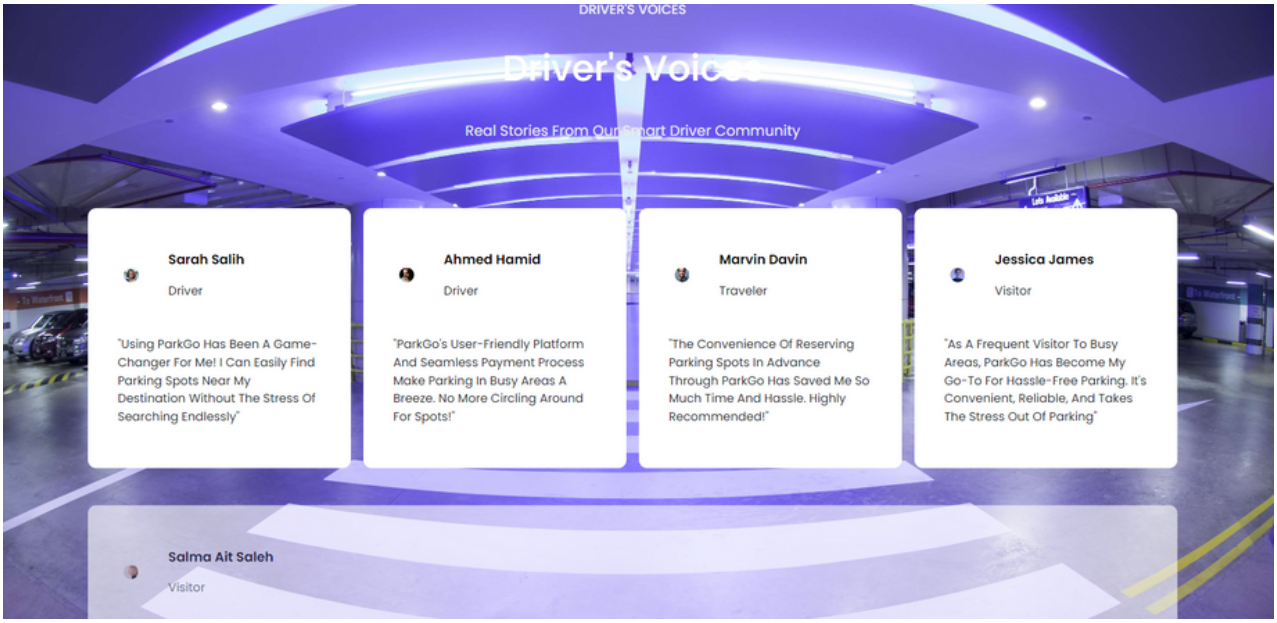
4. Advantages Section:

The advantages section of our website highlights the numerous benefits of utilizing our smart parking system over traditional parking methods. These benefits include time-saving, reduced congestion, environmental sustainability, and enhanced user convenience. By emphasizing the advantages of our smart parking solution, we aim to attract users and encourage them to adopt sustainable transportation practices.



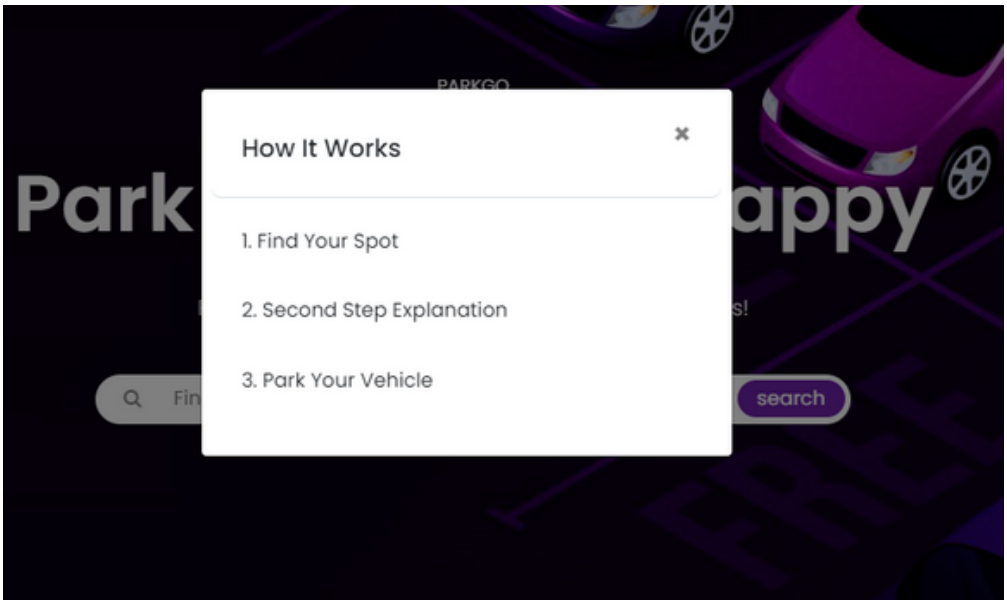
5. Drivers' Voices/Testimonials:

The drivers' voices/testimonials section features feedback and testimonials from actual users of our smart parking system. These testimonials showcase the positive experiences of users and highlight the efficiency and convenience of our platform. Positive user feedback builds trust and credibility, encouraging more users to utilize our smart parking solution.



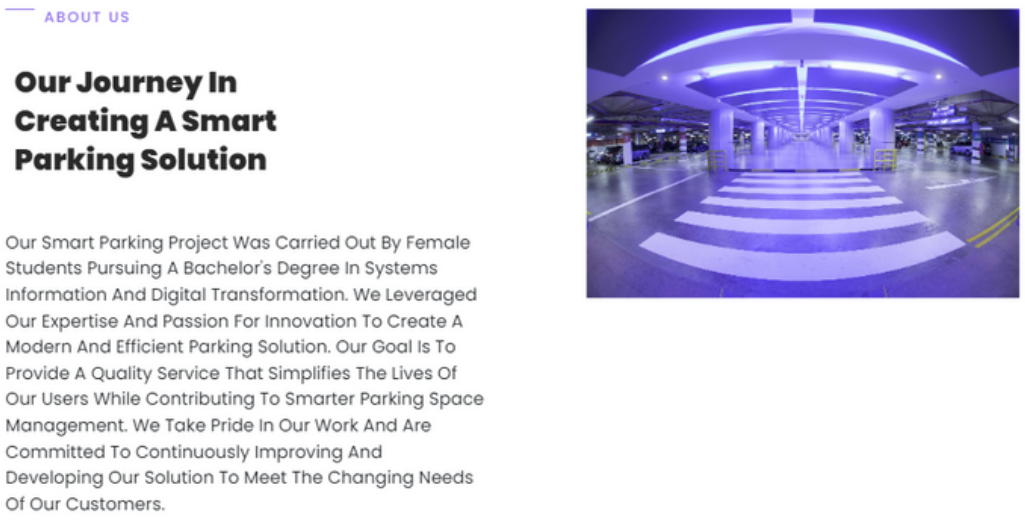
6. How It Works Section:

The "How It Works" section provides a step-by-step explanation of how users can utilize our smart parking system. This section guides users through the process of finding an available parking spot, making a reservation, and completing the parking process. By providing clear instructions and visual aids, we ensure that users can easily navigate and utilize our platform.



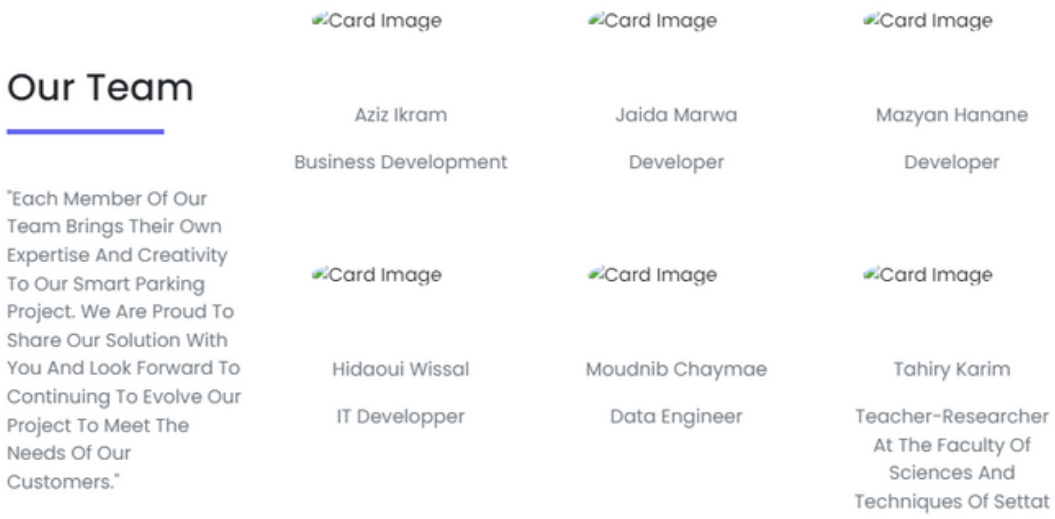
7. Journey of Creation of the Smart Parking:

The journey of creation of our smart parking project documents the development process, challenges faced, innovative solutions implemented, and key milestones achieved. From conceptualization to implementation, our team has worked diligently to bring our vision of a smart parking solution to life. Through collaboration, dedication, and innovation, we have overcome challenges and achieved significant progress in creating a sustainable and efficient parking system.



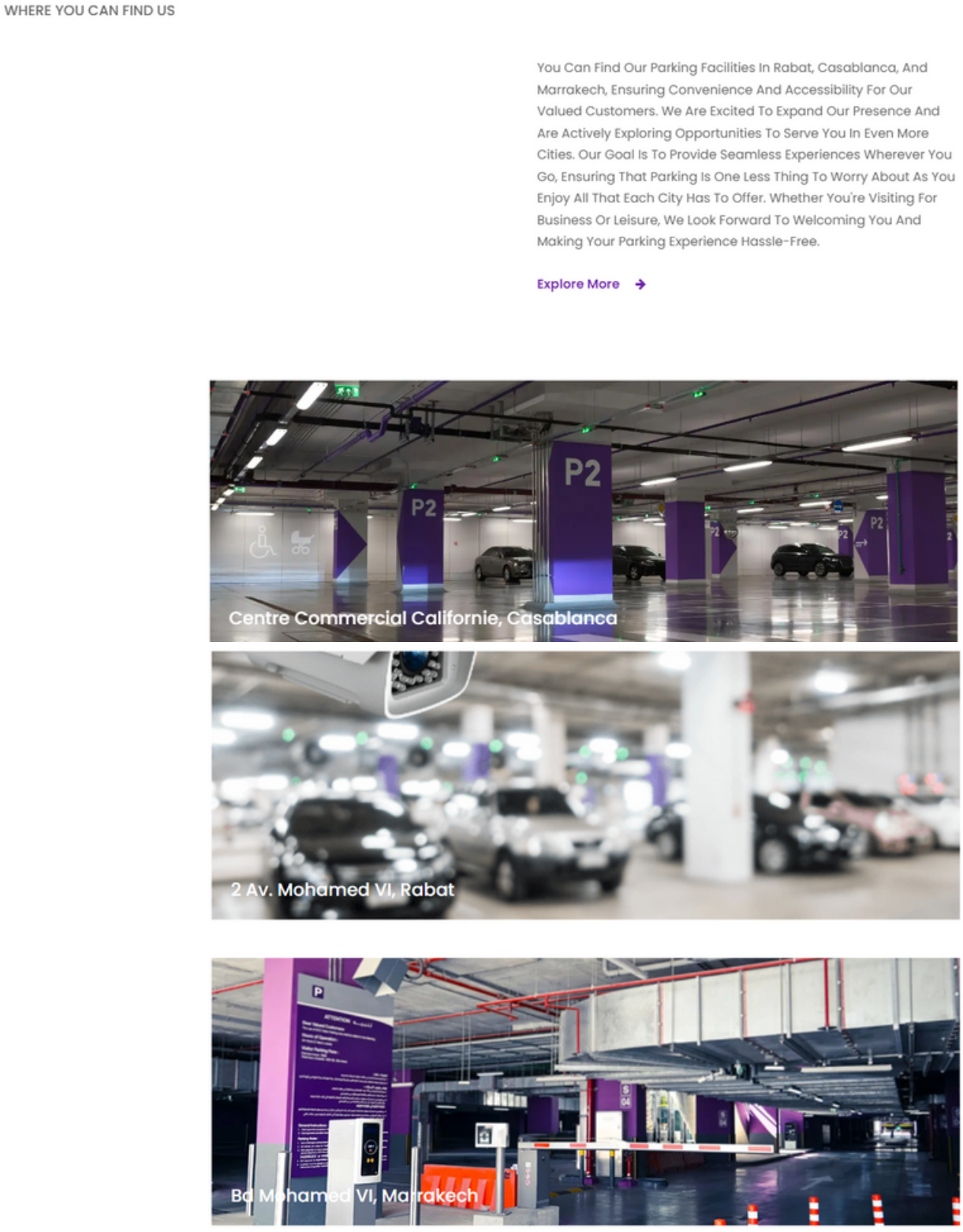
8. Our Team:

The "Our Team" section introduces the members of our team who have contributed to the development and implementation of our smart parking project. Each team member brings unique expertise, skills, and contributions to the project's success. By highlighting the capabilities and achievements of our team, we demonstrate our commitment to delivering high-quality solutions to our users.



9. Locations of Parking Spots:

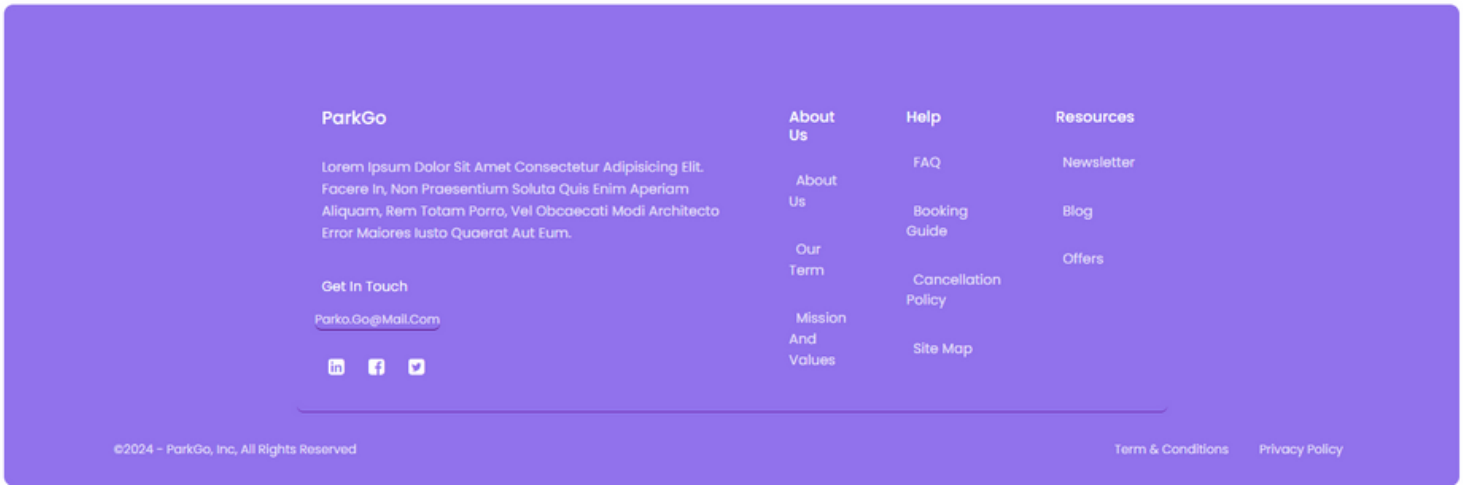
The locations of parking spots section provides information about the availability and accessibility of parking spots covered by our smart parking system in different cities in Morocco. Users can easily locate nearby parking spots and access relevant information such as parking rates, hours of operation, and available amenities. By offering comprehensive information about parking locations, we aim to facilitate user decision-making and enhance the overall user experience.



Footer:

In the footer section of our website, users can access additional resources and information, including:

- About Us: Provides an overview of our company and mission, highlighting our commitment to providing innovative smart parking solutions.
- Help: Offers support and assistance to users who may have questions or encounter issues while using our platform.
- Resources: Provides access to additional resources, such as FAQs, user guides, and tutorials, to help users maximize their experience with our smart parking system.



Blynk:

Incorporating Blynk into our smart parking project has significantly enhanced its functionality and user experience. With the integration of Blynk, we are able to remotely monitor the availability of parking spots in real-time using a user-friendly mobile application. The green LEDs indicate available parking spots, while the red LEDs signify occupied spots, providing a quick and intuitive visual representation of parking availability. This feature not only improves convenience for drivers searching for parking but also contributes to efficient parking management by enabling parking operators to monitor occupancy levels remotely. Additionally, the Blynk platform offers flexibility for future expansions and enhancements, such as implementing advanced analytics and notifications to further optimize parking operations. Overall, the integration of Blynk elevates our smart parking solution, making it more accessible, responsive, and scalable to meet the evolving needs of urban mobility.

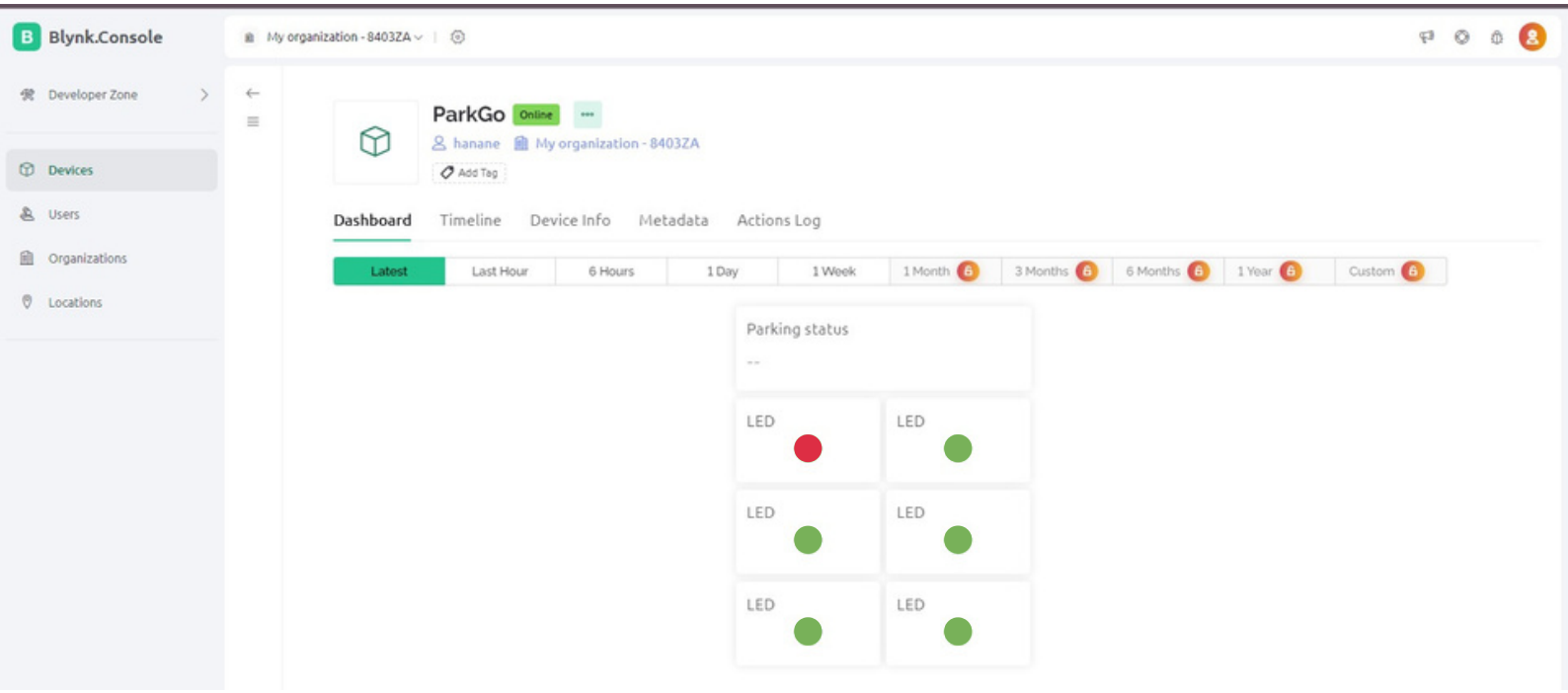


Figure : Screen de notre compte Blynk

Conclusion:

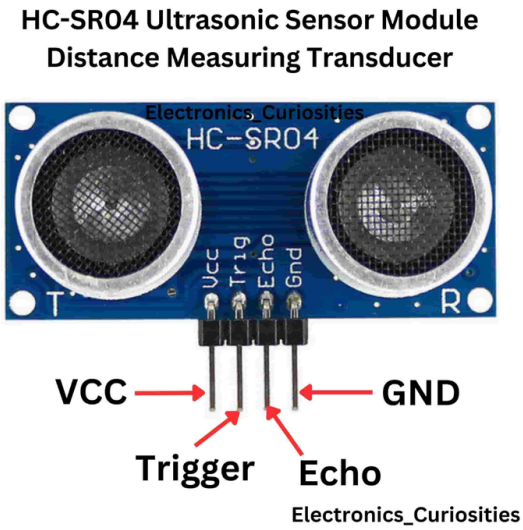
In conclusion, the integration of computer tools such as Arduino and website development platforms has been instrumental in the successful implementation of our smart parking project. Arduino microcontrollers provided a flexible and reliable hardware framework, enabling seamless interfacing with sensors and actuators for efficient car detection and barrier control. The versatility of Arduino allowed us to tailor the hardware components to our specific requirements, ensuring optimal performance and reliability in real-world conditions.

On the software front, the development of a linked website served as the primary interface for users to interact with our smart parking system. Leveraging website development tools and technologies, we created an intuitive and user-friendly platform that facilitated user registration, parking spot reservation, and access to pertinent information. The website's design and functionality were carefully crafted to enhance user experience and streamline the parking process, contributing to improved efficiency and convenience for all stakeholders involved.

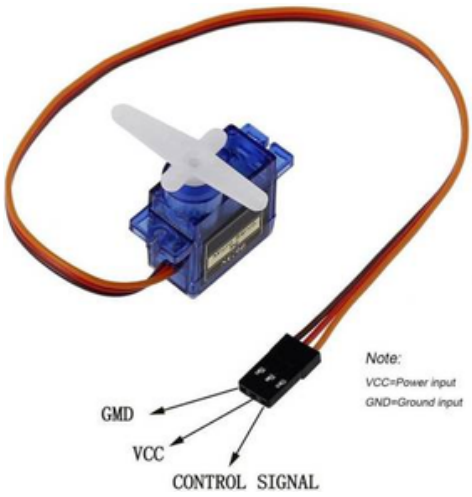
CHAPTER 4: COMPONENT ANALYSIS AND EXPERIMENTAL RESULTS

1. Hardware Requirements

Ultrasonic Sensor: A device that emits ultrasonic waves and measures the time taken for the waves to bounce back after hitting an object. This information is used to detect the presence of nearby objects, such as vehicles in parking spaces.



Servo Motor: A motor that can rotate to a specific angle with high precision. Servo motors are commonly used in applications where precise control of angular position is required, such as opening and closing mechanisms.



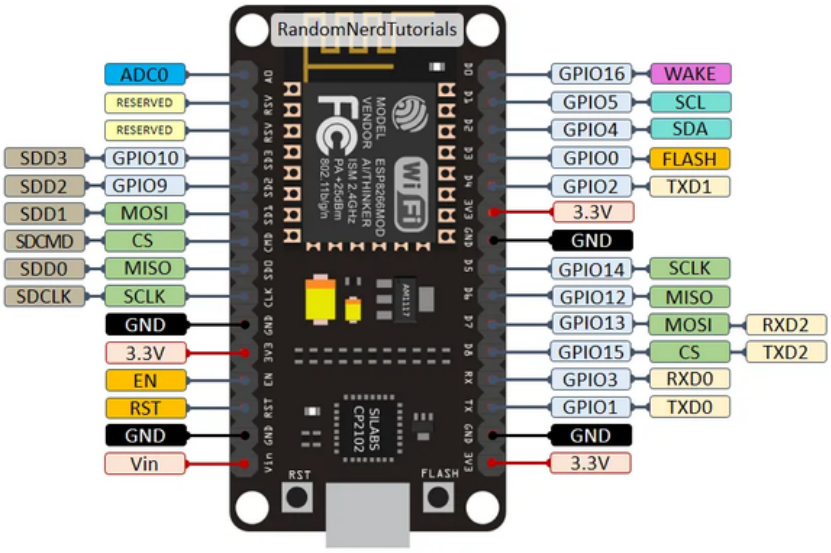
LEDs (Light-Emitting Diodes): Semiconductor devices that emit light when an electric current passes through them. LEDs are commonly used as indicators or visual displays in various electronic devices.



Resistors: Passive electronic components that limit the flow of electric current in a circuit. They are commonly used to control the amount of current flowing to other components, such as LEDs, to prevent damage due to excessive current.



ESP8266: A low-cost, low-power system-on-a-chip (SoC) microcontroller with integrated Wi-Fi and Bluetooth capabilities. It is commonly used in IoT (Internet of Things) applications for wireless communication and data processing.



Cables: Electrical conductors used to connect various components within the circuit. Cables facilitate the transmission of electric signals and power between components, enabling the functioning of the overall system.



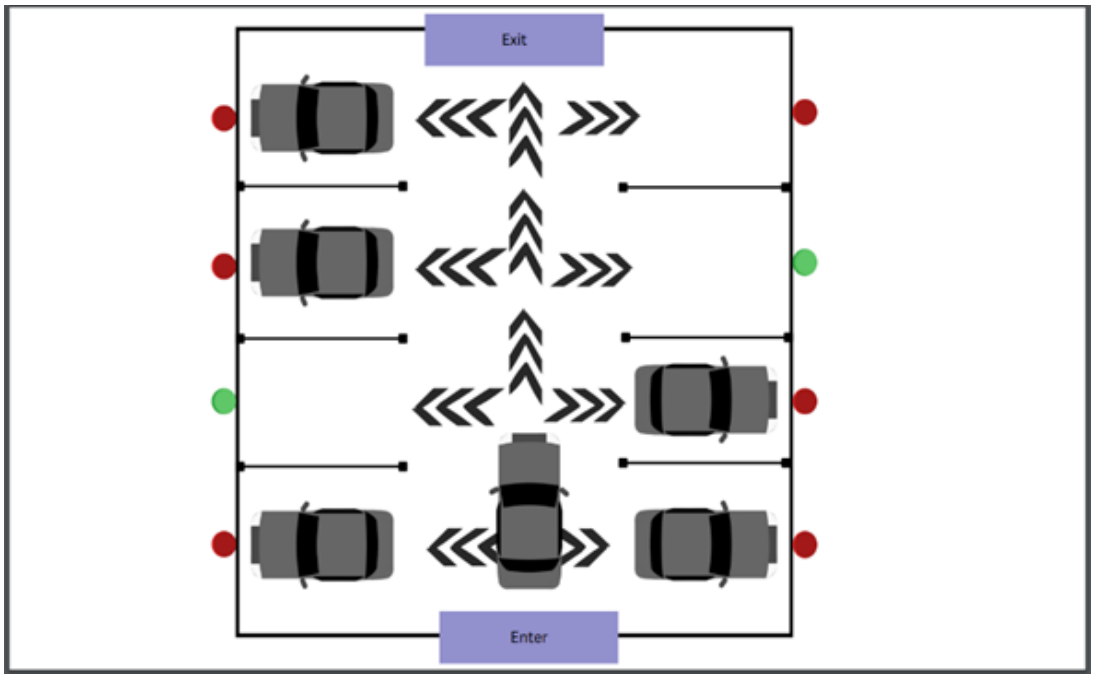
USB Cable: Universal Serial Bus cables used for connecting the ESP8266 microcontroller to power sources or for data transfer purposes. USB cables are commonly used for interfacing with computers or power adapters in electronic projects.



Mini Cars: Small-scale models of cars used for testing and demonstrating the functionality of the smart parking system. Mini cars simulate real vehicles in parking spaces, allowing for the validation of vehicle detection and barrier control mechanisms.

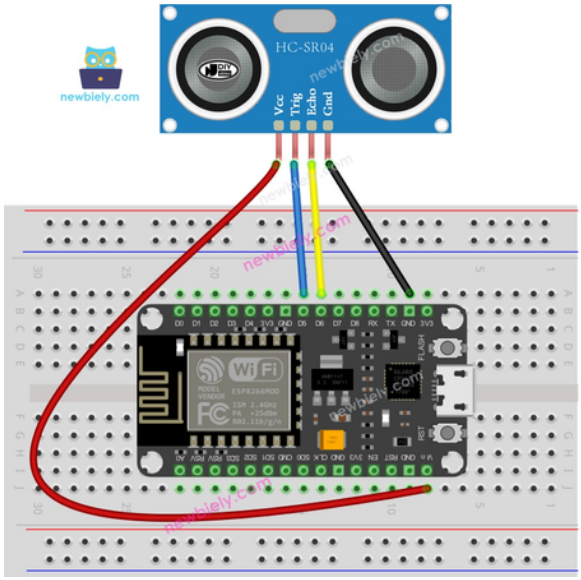


Prototype of the parking maquette: A scaled-down model or replica of the parking area used for planning and visualizing the layout of the smart parking system. The maquette helps in understanding the arrangement of parking spaces, barriers, and other components before actual implementation, facilitating effective design and optimization.

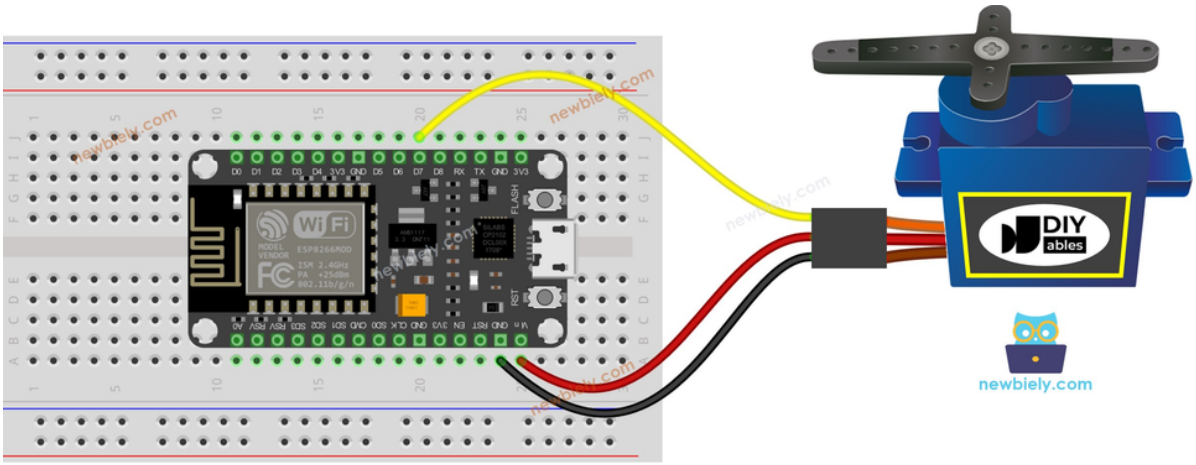


2.Wiring and Assembly of the Implementation:

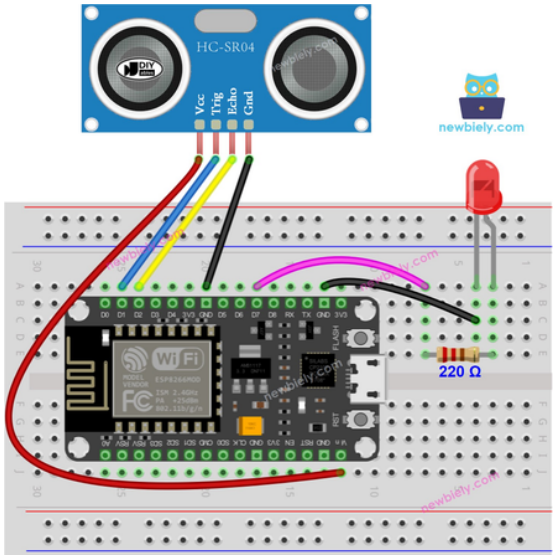
a.Wiring of the Ultrasonic Sensor HC-SR04:



b.Wiring of the Servo Motor:



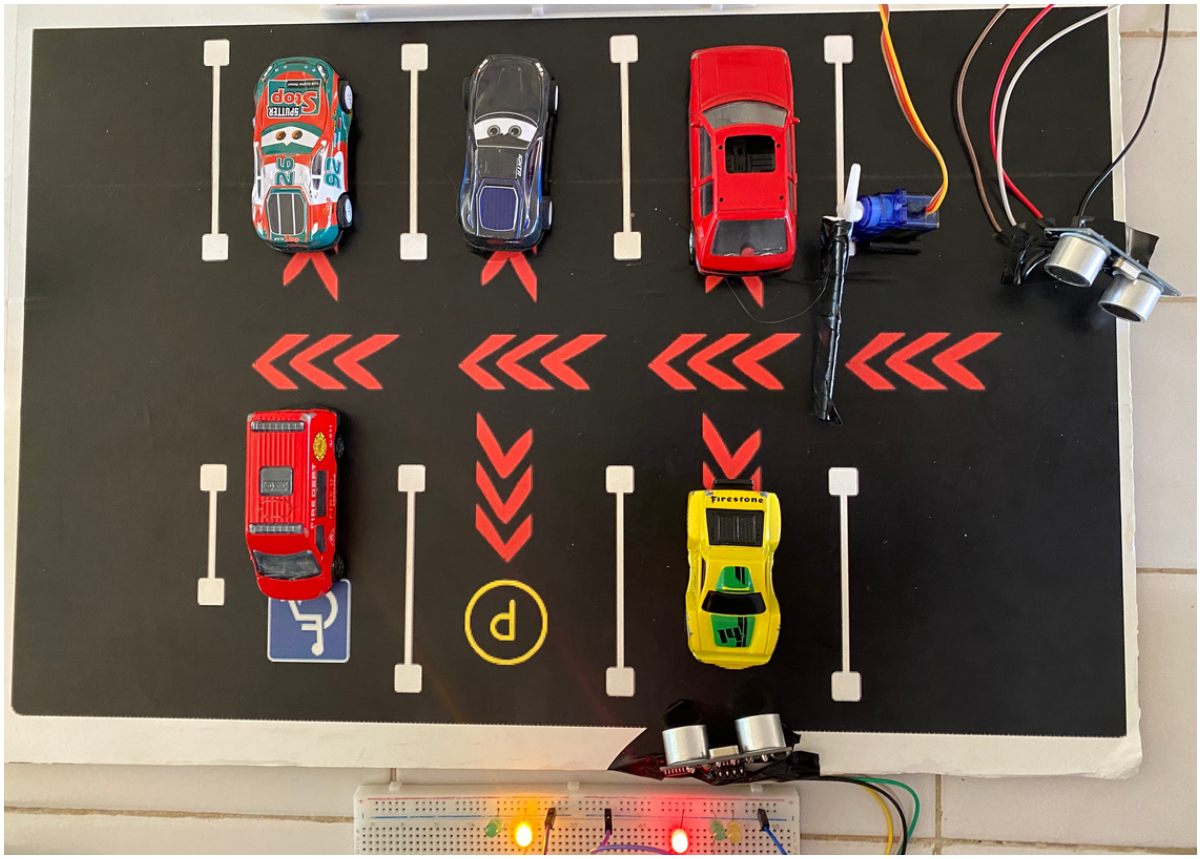
b.Wiring of the leds with the utrasonic sensor



3.Final Result of Our Application:

Our application provides an efficient and user-friendly solution for managing parking spaces in urban environments. By integrating ultrasonic sensors, servo motors, LEDs, and an ESP32 microcontroller, we have created a smart parking system that accurately detects the presence of vehicles, controls parking access barriers, and provides real-time status updates to users. With the help of a visually intuitive website interface, drivers can easily locate available parking spaces and reserve them in advance, reducing congestion and optimizing parking space utilization. The final result is a comprehensive smart parking solution that enhances the parking experience for both drivers and administrators, contributing to improved traffic flow and urban mobility.

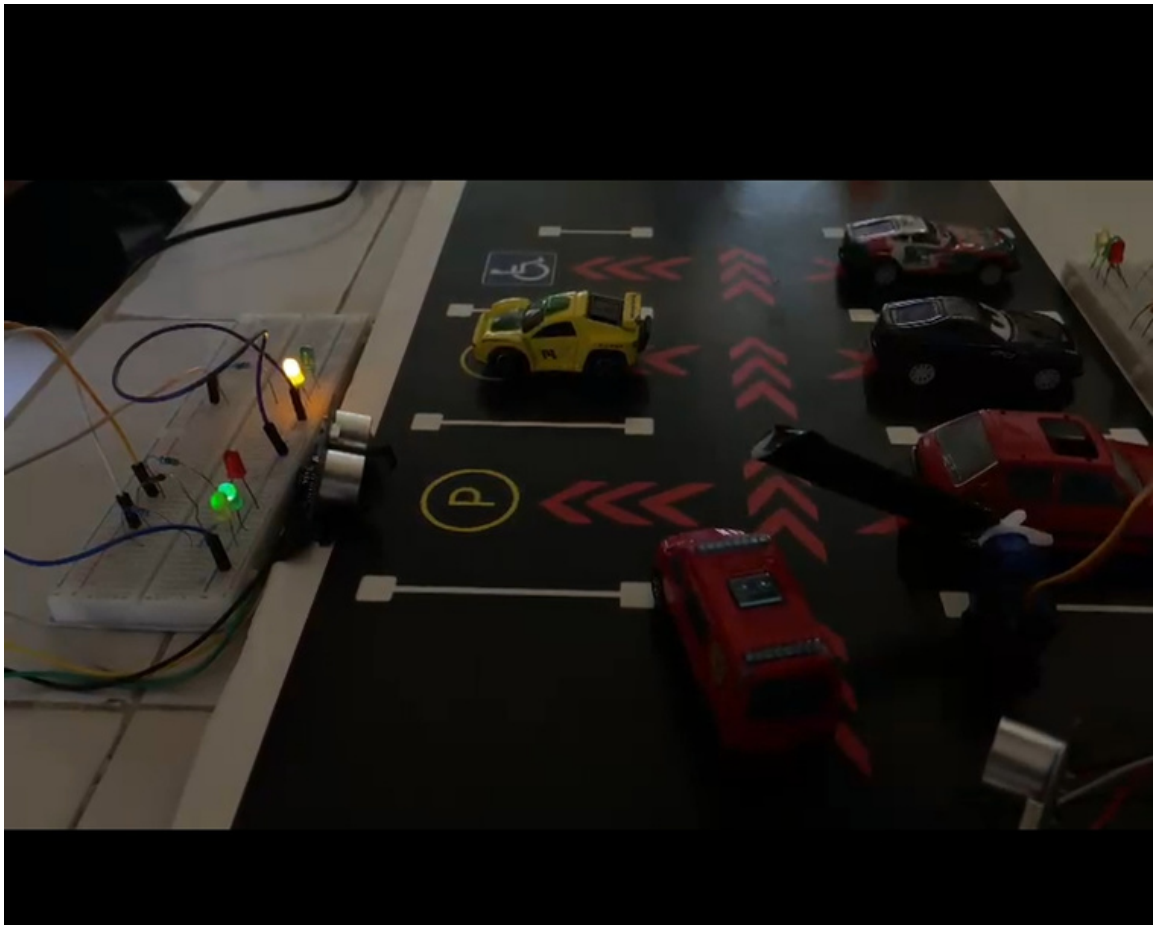
Overall results:



The barrier is closed (No car is near by 5 centimeters)



Available spot and reserved one:



The available spot becomes full after parking the car:



CONCLUSION:

In conclusion, our smart parking project represents a significant advancement in urban infrastructure management and technology integration. By combining hardware components such as ultrasonic sensors and servo motors with software elements like the ESP8266 microcontroller and website interface, we have developed a comprehensive solution that addresses the challenges of parking space management in urban areas.



Our smart parking system, after thorough testing and refinement, has proven its reliability in detecting vehicles, managing access, and providing real-time updates. With a user-friendly website interface, drivers can effortlessly find and reserve parking spaces, leading to reduced congestion and improved urban mobility.



Looking ahead, the success of our project opens up new possibilities for further innovation and expansion. By leveraging emerging technologies and continuous feedback from users, we can continue to improve and refine our smart parking system to meet the evolving needs of urban environments.



Overall, our smart parking project exemplifies the transformative potential of technology in addressing real-world challenges and improving the quality of life for communities. Through collaboration, innovation, and a commitment to sustainability, we are paving the way towards smarter and more efficient urban spaces.